The MAID Model

Multi-Agent Impersonation with Fuzzy Knowledge

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Abstract

We introduce the **MAID model** (Multi-Agent Impersonation with Fuzzy Knowledge), a game-theoretic framework for a heterogeneous economy populated by a domestic representative agent D, an international representative agent I, and a common adversary A. Each agent chooses an impersonation level $j \in [0, 1]$ and possesses fuzzy knowledge $K_j \in [0, 1]$ of a shared market M. We derive the unique Nash equilibrium in closed form, characterise its comparative-static properties, establish existence and uniqueness conditions, and analyse welfare and collapse dynamics. The adversary is shown to introduce a *knowledge wedge* that suppresses cooperative market participation by D and I even when the adversary's own knowledge is low. We complement the analytical results with an algorithm for computing the equilibrium under parameter uncertainty and discuss connections to robust control, mechanism design, and adversarial machine learning.

The paper ends with "The End"

Keywords: impersonation games, fuzzy knowledge, Nash equilibrium, adversarial agents, heterogeneous economy, robust mechanism design.

Contents

1 Introduction **3**

2 The Economic Environment **3**
2.1 Agents and action spaces . . . 3
2.2 The market signal . . . 3
2.3 Payoff structure . . . 4

3 Nash Equilibrium **4**
3.1 Best-response functions . . . 4
3.2 Linear system . . . 4
3.3 Closed-form solution . . . 5
3.4 Symmetric equilibrium . . . 5

4 Theoretical Properties **5**
4.1 The adversary's knowledge wedge . . . 5
4.2 Strategic substitutability . . . 5
4.3 Collapse threshold . . . 6
4.4 Existence and uniqueness summary . . . 6

5 Statistical Foundations **6**
5.1 Bayesian updating of fuzzy knowledge . . . 6
5.2 Variance of equilibrium impersonation . . . 6
5.3 Moment conditions for GMM estimation . . . 6

6 Financial Applications **7**
6.1 Market microstructure interpretation . . . 7
6.2 Value-at-Risk under impersonation . . . 7

7 Algorithm for Equilibrium Computation **7**
7.1 Projected gradient descent . . . 7
7.2 Complexity . . . 8

8 Connections to AI and Machine Learning **8**
8.1 Adversarial learning interpretation . . . 8
8.2 Fuzzy knowledge as epistemic uncertainty . . . 8
8.3 Multi-agent reinforcement learning . . . 8

9 Welfare Analysis and Mechanism Design **8**
9.1 Social surplus . . . 8
9.2 Price of anarchy . . . 9

10 Graphical Analysis **9**
10.1 Agent interaction diagram . . . 9
10.2 Best-response functions . . . 9
10.3 Equilibrium surface . . . 9

11 Discussion **9**

12 Conclusion **10**

List of Figures

1 Three-agent MAID interaction structure. Arrows represent market participation weighted by fuzzy knowledge and impersonation level. Double-headed arrow between D and I indicates mutual interference. The adversary A introduces one-sided interference on both D and I. . . . 9
2 Best-response function $x_D^*(K_D)$ for $K_I = 0.6$, $K_A = 0.4$, $\mu/\gamma = 1$. Higher interference ratio ϱ steepens the slope and shifts the zero-impersonation threshold rightward. . . . 10
3 Equilibrium surface $x_D^*(K_D, \varrho)$ with $K_I = K_A = 0.5$. The surface flattens as $\varrho \to 1$ (collapse) and rises steeply with K_D when ϱ is small. . . . 10

List of Algorithms

1 Projected Gradient Descent for MAID Equilibrium . . . 7

List of Tables

1 Model parameters and domains . . . 4
2 Equilibrium regimes by interference ratio $\varrho = \phi/\gamma$ . . . 6
3 Value-at-Risk under representative parameterisations ($\mu = 1$, $\gamma = 1$, $\phi = 0.3$) . . . 7

1 Introduction

Modern financial markets and information economies are populated by agents who simultaneously participate in genuine trade and engage in strategic misrepresentation—*impersonation*—of their identities, intentions, or beliefs. Central banks, sovereign wealth funds, and algorithmic traders all operate in environments where adversaries may deliberately inject false signals, and where no agent possesses complete knowledge of the underlying market mechanism.

The *MAID model* formalises this environment. Three representative agents—domestic (D), international (I), and adversarial (A)—interact over a common market M. Each agent selects an impersonation intensity $j \in [0,1]$ and holds *fuzzy knowledge* $K_j \in [0,1]$, a scalar capturing the fraction of the true market signal the agent can observe without noise. The model nests standard information-asymmetry frameworks as special cases and admits a tractable closed-form equilibrium.

Contributions.

(i) We derive the unique interior Nash equilibrium in closed form (Section 3) and prove existence and uniqueness under mild regularity conditions (Section 4).

(ii) We characterise the adversary's *knowledge wedge*: even a low-knowledge adversary suppresses equilibrium participation by D and I (Proposition 4.1).

(iii) We identify a *collapse threshold* at which mutual interference dominates and all impersonation collapses to zero (Theorem 4.4).

(iv) We provide an algorithm for computing the equilibrium under parameter uncertainty via projected gradient descent (Section 7).

(v) We establish welfare implications and mechanism-design corollaries (Section 9).

Related literature. The MAID model draws on four strands of literature. Strategic information transmission originates with [1] and [5]. Fuzzy-set approaches to bounded rationality follow [11] and [2]. Adversarial game theory connects to [9] and the robust-control literature of [6]. Adversarial machine learning—where an agent perturbs inputs to fool a model—is surveyed in [4].

2 The Economic Environment

2.1 Agents and action spaces

The economy consists of three representative agents indexed by $j \in \mathcal{J} = \{D, I, A\}$.

Definition 2.1 (Impersonation level)**.** Agent j's *impersonation level* is a scalar $x_j \in [0,1]$, where $x_j = 0$ denotes full authenticity and $x_j = 1$ denotes complete impersonation. The action profile is $\mathbf{x} = (x_D, x_I, x_A) \in [0,1]^3$.

Definition 2.2 (Fuzzy knowledge)**.** Agent j's *fuzzy knowledge* of market M is a scalar $K_j \in [0,1]$, interpreted as a possibility measure in the sense of [2]: $K_j = 1$ corresponds to full knowledge and $K_j = 0$ to complete ignorance.

2.2 The market signal

Let $M \in \mathbb{R}_+$ denote the true market value. Each agent observes a noisy signal:

$$\tilde{M}_j \;=\; K_j\, M \;+\; (1 - K_j)\, \varepsilon_j, \qquad \varepsilon_j \sim U[0,1], \tag{1}$$

where the ε_j are independent. Taking expectations, $\mathbb{E}[\tilde{M}_j] = K_j M + (1 - K_j)/2$, so higher knowledge shifts the signal toward the true value. The conditional variance is $(1 - K_j)^2/12$.

Assumption 2.3 (Signal independence)**.** The noise terms ε_D, ε_I, ε_A are mutually independent and independent of M.

2.3 Payoff structure

Agent j's payoff from profile $(x_j, \mathbf{x}_{-j})$ is

$$\Pi_j(x_j, \mathbf{x}_{-j}) \;=\; \mu\, K_j\, \tilde{M}_j \;-\; \frac{\gamma}{2}\, x_j^2 \;-\; \phi\, x_j \sum_{k \neq j} x_k, \tag{2}$$

where $\mu > 0$ is the marginal market gain, $\gamma > 0$ is the convex cost coefficient of impersonation, and $\phi \geq 0$ is the bilateral interference parameter.
The three terms in (2) have natural economic interpretations:

- $\mu K_j \tilde{M}_j$: the revenue from exploiting the market signal, scaled by knowledge quality.
- $\frac{\gamma}{2} x_j^2$: the strictly convex private cost of sustaining impersonation (reputational risk, operational complexity).
- $\phi\, x_j \sum_{k \neq j} x_k$: the cross-impersonation penalty capturing mutual interference when multiple agents simultaneously impersonate.

Table 1 summarises all parameters and their domains.

Table 1: Model parameters and domains

Symbol	Description	Domain	Sign
K_j	Fuzzy knowledge of agent j	$[0, 1]$	$+$
x_j	Impersonation level of agent j	$[0, 1]$	$+$
M	True market value	$\mathbb{R}_+$	$+$
μ	Marginal market gain	$\mathbb{R}_{++}$	$+$
γ	Cost coefficient	$\mathbb{R}_{++}$	$+$
ϕ	Interference coefficient	$\mathbb{R}_+$	$+$
ϱ	Interference ratio ϕ/γ	$[0, \infty)$	$+$

3 Nash Equilibrium

3.1 Best-response functions

Taking the first-order condition of (2) with respect to x_j:

$$\frac{\partial \Pi_j}{\partial x_j} \;=\; \mu K_j \;-\; \gamma x_j \;-\; \phi \sum_{k \neq j} x_k \;=\; 0. \tag{3}$$

The second-order condition $\partial^2 \Pi_j / \partial x_j^2 = -\gamma < 0$ is satisfied globally, confirming strict concavity. The unconstrained best response is

$$\mathrm{BR}_j(\mathbf{x}_{-j}) \;=\; \frac{\mu K_j - \phi \sum_{k \neq j} x_k}{\gamma}. \tag{4}$$

3.2 Linear system

Collecting the three FOCs (3) yields the linear system

$$\underbrace{\begin{pmatrix} \gamma & \phi & \phi \\ \phi & \gamma & \phi \\ \phi & \phi & \gamma \end{pmatrix}}_{\boldsymbol{\Gamma}} \begin{pmatrix} x_D \\ x_I \\ x_A \end{pmatrix} \;=\; \mu \begin{pmatrix} K_D \\ K_I \\ K_A \end{pmatrix}. \tag{5}$$

Writing $\boldsymbol{\Gamma} = \gamma(\mathbf{I} + \varrho\, \mathbf{W})$ where $\varrho = \phi/\gamma$ and $\mathbf{W}$ is the 3×3 matrix of ones minus the identity, (5) becomes

$$(\mathbf{I} + \varrho\, \mathbf{W})\, \mathbf{x}^* \;=\; \frac{\mu}{\gamma}\, \mathbf{K}. \tag{6}$$

3.3 Closed-form solution

Theorem 3.1 (Nash Equilibrium). *Under Assumption 2.3, if $\varrho \neq 1$ and $\varrho \neq -\frac{1}{2}$, the game has a unique Nash equilibrium given by*

$$x_j^* \;=\; \frac{\mu}{\gamma} \cdot \frac{K_j(1+\varrho) - \varrho(K_{-j,1} + K_{-j,2})}{(1-\varrho)(1+2\varrho)}, \qquad j \in \{D, I, A\}, \tag{7}$$

where $(K_{-j,1}, K_{-j,2})$ denotes the knowledge levels of the two agents other than j, and $\varrho = \phi/\gamma$. The constrained equilibrium imposes $x_j^* \leftarrow \max(0, \min(1, x_j^*))$.

Proof. The matrix $(\mathbf{I} + \varrho\mathbf{W})$ has eigenvalues $\lambda_1 = 1 - \varrho$ (with multiplicity 2, eigenvectors orthogonal to $\mathbf{1}$) and $\lambda_2 = 1 + 2\varrho$ (with multiplicity 1, eigenvector $\mathbf{1}/\sqrt{3}$). Both eigenvalues are non-zero under the stated conditions, so $(\mathbf{I} + \varrho\mathbf{W})$ is invertible. Its inverse is

$$(\mathbf{I} + \varrho\mathbf{W})^{-1} \;=\; \frac{1}{1-\varrho}\mathbf{I} \;-\; \frac{\varrho}{(1-\varrho)(1+2\varrho)}\mathbf{W}_{\text{ones}}, \tag{8}$$

where $\mathbf{W}_{\text{ones}}$ is the all-ones matrix of size 3. Multiplying both sides of (6) by $(\mathbf{I} + \varrho\mathbf{W})^{-1}$ and collecting terms yields (7). Uniqueness follows from strict concavity of each payoff and linearity of the FOC system. □

3.4 Symmetric equilibrium

If $K_D = K_I = K_A = K$, equation (7) reduces to

$$x^* \;=\; \frac{\mu\, K}{\gamma(1+2\varrho)}, \tag{9}$$

which is strictly decreasing in ϱ. Higher mutual interference uniformly suppresses impersonation.

4 Theoretical Properties

4.1 The adversary's knowledge wedge

Proposition 4.1 (Knowledge wedge). *Agent D's equilibrium impersonation x_D^* is strictly decreasing in both x_I^* and x_A^*. In particular, the adversary's equilibrium action reduces D's payoff by $\phi\, x_A^*\, x_D^*$ and suppresses x_D^* by*

$$\frac{\partial x_D^*}{\partial K_A} \;=\; -\frac{\mu\,\varrho}{\gamma(1-\varrho)(1+2\varrho)} \;<\; 0 \quad \textit{for } \varrho > 0. \tag{10}$$

Proof. Differentiating (7) with respect to K_A and using the chain rule gives (10) directly. The sign follows from $\varrho > 0$ and the denominator being positive in the interior regime. □

Remark 4.2. Proposition 4.1 formalises the *adversarial crowding-out* effect: a knowledge-poor adversary ($K_A \to 0$) still suppresses D and I through the interference term ϕ, even when its own equilibrium impersonation x_A^* is negligible.

4.2 Strategic substitutability

Proposition 4.3 (Strategic substitutability). *Impersonation levels are strategic substitutes: for all $j \neq k$,*

$$\frac{\partial \mathrm{BR}_j}{\partial x_k} \;=\; -\frac{\phi}{\gamma} \;=\; -\varrho \;\leq\; 0. \tag{11}$$

The cross-knowledge comparative static satisfies

$$\frac{\partial x_j^*}{\partial K_j} \;=\; \frac{\mu(1+\varrho)}{\gamma(1-\varrho)(1+2\varrho)} \;>\; 0, \qquad \frac{\partial x_j^*}{\partial K_k} \;=\; -\frac{\mu\,\varrho}{\gamma(1-\varrho)(1+2\varrho)} \;\leq\; 0, \quad k \neq j. \tag{12}$$

Proof. Equation (11) follows immediately from (4). Equations (12) follow by differentiating (7). □

4.3 Collapse threshold

Theorem 4.4 (Collapse)**.** *As $\varrho = \phi/\gamma \to 1^-$, the unconstrained equilibrium diverges, and the unique constrained equilibrium collapses to the zero vector: $\mathbf{x}^* \to \mathbf{0}$. More precisely, for the symmetric case (9), x^* is maximised at $\varrho = 0$ and equals zero at $\varrho \to \infty$. For the asymmetric case, if $\varrho \to 1$ and K_j are not all equal, the system loses the unique interior equilibrium and at least one agent is at a corner solution.*

Proof. When $\varrho \to 1$, the eigenvalue $\lambda_1 = 1 - \varrho \to 0$, so $(\mathbf{I} + \varrho\mathbf{W})^{-1}$ is unbounded. The numerator in (7) remains bounded (it depends only on $K_j \in [0, 1]$), while the denominator $(1 - \varrho)(1 + 2\varrho) \to 0$. Hence the unconstrained solution diverges. The constraint $x_j^* \in [0, 1]$ then forces at least one binding corner. In the symmetric case, $x^* \to 0$ as $1 + 2\varrho \to \infty$. □

4.4 Existence and uniqueness summary

Table 2 summarises the equilibrium regimes as functions of ϱ.

Table 2: Equilibrium regimes by interference ratio $\varrho = \phi/\gamma$

Regime	Condition	Equilibrium	Economic interpretation
Interior	$\varrho \in [0, 1) \setminus \{-\frac{1}{2}\}$	Unique, interior	Normal trading with impersonation
Corner	$x_j^* \notin (0, 1)$ for some j	Constrained	One or more agents at boundary
Bifurcation	$\varrho = 1$	None (diverges)	Collapse; coordination failure
Negative-ϱ	$\varrho \in (-\frac{1}{2}, 0)$	Unique, interior	Complementarity; over-impersonation

5 Statistical Foundations

5.1 Bayesian updating of fuzzy knowledge

Agent j updates K_j upon observing a private signal s_j. Under a Beta prior $K_j \sim \text{Beta}(\alpha_j, \beta_j)$ and a Binomial likelihood, the posterior is

$$K_j \mid s_j \;\sim\; \text{Beta}(\alpha_j + s_j,\; \beta_j + n_j - s_j)\,, \tag{13}$$

where n_j is the number of market observations and s_j is the number of correct predictions. The posterior mean $\hat{K}_j = (\alpha_j + s_j)/(\alpha_j + \beta_j + n_j)$ serves as the equilibrium input.

5.2 Variance of equilibrium impersonation

Treating K_j as random with mean $\bar{K}_j$ and variance $\sigma^2_{K_j}$, a first-order delta method expansion of (7) gives

$$\text{Var}(x_j^*) \;\approx\; \left(\frac{\mu(1+\varrho)}{\gamma(1-\varrho)(1+2\varrho)}\right)^2 \sigma^2_{K_j} \;+\; \left(\frac{\mu\,\varrho}{\gamma(1-\varrho)(1+2\varrho)}\right)^2 \sum_{k \neq j} \sigma^2_{K_k}, \tag{14}$$

showing that equilibrium variance increases in ϱ: higher interference amplifies knowledge uncertainty.

5.3 Moment conditions for GMM estimation

Suppose we observe T repetitions of the market interaction. Define $e_{jt} = x_{jt} - x_j^*(\boldsymbol{\theta})$ where $\boldsymbol{\theta} = (\mu, \gamma, \phi)^\top$. The GMM estimator minimises

$$\hat{\boldsymbol{\theta}} \;=\; \arg\min_{\boldsymbol{\theta}} \left[\frac{1}{T}\sum_{t=1}^{T} \mathbf{e}_t\right]^\top \mathbf{W}_T \left[\frac{1}{T}\sum_{t=1}^{T} \mathbf{e}_t\right], \tag{15}$$

where $\mathbf{W}_T$ is the efficient weighting matrix. Under standard regularity, $\sqrt{T}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \mathbf{V})$.

6 Financial Applications

6.1 Market microstructure interpretation

In a Kyle (1985)-style market, the *price impact* of aggregate order flow $Q = x_D + x_I$ is λQ, where $\lambda > 0$ is market depth. The adversary A submits noise trades that inflate Q, widening the bid–ask spread. Mapping $\phi \leftrightarrow \lambda$ and $\mu \leftrightarrow$ (expected profit per unit), the MAID equilibrium (7) recovers the Kyle insider's optimal quantity as a special case when $K_A = 0$.

6.2 Value-at-Risk under impersonation

Let portfolio return $R_j = x_j^* \tilde{M}_j - \frac{\gamma}{2}(x_j^*)^2$. Using (1), $\mathrm{Var}(R_j) = \mu^2 (x_j^*)^2 (1 - K_j)^2/12$. The 99% Value-at-Risk is

$$\mathrm{VaR}_{0.99}^{(j)} \;=\; -\mathbb{E}[R_j] \;+\; 2.576\,\mu\, x_j^* \,\frac{1 - K_j}{2\sqrt{3}}, \tag{16}$$

which is increasing in $1 - K_j$ (ignorance) and increasing in x_j^*. Table 3 reports indicative values.

Table 3: Value-at-Risk under representative parameterisations ($\mu = 1$, $\gamma = 1$, $\phi = 0.3$)

Agent	K_j	x_j^*	$\mathbb{E}[R_j]$	$\mathrm{VaR}_{0.99}^{(j)}$
D	0.70	0.558	0.154	0.2388
I	0.60	0.429	0.087	0.2003
A	0.40	0.171	0.015	0.1097

7 Algorithm for Equilibrium Computation

7.1 Projected gradient descent

When parameters are estimated with uncertainty, the closed-form solution (7) may violate the box constraint $[0, 1]^3$. Algorithm 1 computes the constrained equilibrium via projected gradient descent on the potential function $\Phi(\mathbf{x}) = -\sum_j \Pi_j(\mathbf{x})$.

Algorithm 1 Projected Gradient Descent for MAID Equilibrium

Require: Parameters $\mu, \gamma, \phi > 0$; knowledge vector $\mathbf{K} \in [0, 1]^3$; step size $\eta > 0$; tolerance $\varepsilon > 0$; max iterations T.
Ensure: Approximate Nash equilibrium $\mathbf{x}^* \in [0, 1]^3$.

```
Initialise x^(0) ← K                                   ▷ Knowledge as prior
for t = 0, 1, ..., T − 1 do
    Compute gradient: g_j^(t) ← μK_j − γx_j^(t) − φ Σ_{k≠j} x_k^(t) for all j
    Gradient step: x̃_j ← x_j^(t) + η g_j^(t)
    Project onto [0, 1]: x_j^(t+1) ← max(0, min(1, x̃_j))
    if ‖x^(t+1) − x^(t)‖_∞ < ε then
        return x^(t+1)
    end if
end for
return x^(T)
```

Proposition 7.1 (Convergence of Algorithm 1)**.** *For step size $\eta \in (0, 2/(\gamma + 2\phi))$, Algorithm 1 converges linearly to the unique Nash equilibrium (7) (or its constrained projection) at rate $r = 1 - \eta(\gamma - 2\phi)$.*

Proof. The potential function Φ is $(\gamma - 2\phi)$-strongly convex (since $\gamma > 2\phi$ in the interior regime) and $(\gamma + 2\phi)$-smooth. Standard projected gradient convergence results then apply; see [8]. □

7.2 Complexity

Each iteration of Algorithm 1 costs $O(|\mathcal{J}|^2) = O(9)$ floating-point operations (constant in the three-agent model). The algorithm achieves ε-accuracy in at most $\lceil \log(1/\varepsilon)/\log(1/r) \rceil$ iterations, polynomial in $\log(1/\varepsilon)$.

8 Connections to AI and Machine Learning

8.1 Adversarial learning interpretation

The adversary A in the MAID model is analogous to the attacker in an adversarial machine learning framework [4]. Impersonation level x_A corresponds to the perturbation budget $\delta \in [0, 1]$ under an ℓ_∞ threat model, while fuzzy knowledge K_A proxies the attacker's access to model gradients (white-box: $K_A \to 1$; black-box: $K_A \to 0$).

The defender D's optimal response to adversarial perturbations is the minimax objective

$$\min_{x_D} \max_{x_A} \Pi_D(x_D, x_A, x_I^*) \;=\; \min_{x_D} \left[\mu K_D \tilde{M}_D - \tfrac{\gamma}{2} x_D^2 - \phi\, x_D\, x_A^*(x_D)\right], \tag{17}$$

where $x_A^*(x_D)$ is the adversary's best response. Substituting (4), the inner maximisation yields $x_A^* = (\mu K_A - \phi x_D - \phi x_I^*)/\gamma$, giving a Stackelberg equilibrium in which D is the leader.

8.2 Fuzzy knowledge as epistemic uncertainty

In Bayesian deep learning, epistemic uncertainty is quantified via the posterior variance of model weights. The MAID mapping is: $K_j \leftrightarrow 1 - \sigma_j^2/\sigma_{\max}^2$, where σ_j^2 is the posterior variance. Under this mapping, (14) recovers the variance propagation formula in [3] (Proposition 1).

8.3 Multi-agent reinforcement learning

In a repeated version of the MAID game, agents may learn via Q-learning. The Q-function for agent j at state $\mathbf{x}$ and action a is

$$Q_j(a, \mathbf{x}) \;=\; \Pi_j(a, \mathbf{x}_{-j}) \;+\; \beta\, \mathbb{E}\left[\max_{a'} Q_j(a', \mathbf{x}')\right], \tag{18}$$

where $\beta \in (0, 1)$ is the discount factor and $\mathbf{x}'$ is the next state. The unique Nash equilibrium (7) is the fixed point of the best-response correspondence and is therefore the unique Nash Q-equilibrium under sufficient exploration [9].

9 Welfare Analysis and Mechanism Design

9.1 Social surplus

Define aggregate welfare as $W(\mathbf{x}) = \sum_j \Pi_j(\mathbf{x})$. At the Nash equilibrium,

$$W^* \;=\; \sum_j \left[\mu K_j \tilde{M}_j - \tfrac{\gamma}{2}(x_j^*)^2 - \phi\, x_j^* \sum_{k \neq j} x_k^*\right]. \tag{19}$$

The social optimum $\mathbf{x}^{SO}$ solves $\max_{\mathbf{x}} W(\mathbf{x})$, giving

$$x_j^{SO} \;=\; \frac{\mu K_j}{\gamma + 2\phi}, \tag{20}$$

which is strictly less than x_j^* when $\varrho > 0$ (Proposition 4.3). Hence the Nash equilibrium over-impersonates relative to the social optimum.

Corollary 9.1 (Impersonation tax)**.** *A Pigouvian tax $\tau^* = \phi \sum_{k \neq j} x_k^*$ on agent j's impersonation decentralises the social optimum.*

Proof. Replacing Π_j with $\Pi_j - \tau x_j$ shifts the FOC to $\mu K_j - (\gamma + \tau) x_j - \phi \sum_{k \neq j} x_k = 0$. Setting $\tau = \phi \sum_{k \neq j} x_k^*$ and noting that at $\mathbf{x}^{SO}$ this matches (20) completes the proof. □

9.2 Price of anarchy

The *price of anarchy* (PoA) quantifies welfare loss from non-cooperative play:

$$\text{PoA} \;=\; \frac{W(\mathbf{x}^{SO})}{W(\mathbf{x}^{*})} \;\geq\; 1. \tag{21}$$

For the symmetric case with $K_D = K_I = K_A = K$:

$$\text{PoA} \;=\; \frac{(\gamma+2\phi)^2}{(1+2\varrho)^2\,\gamma^2/(1+2\varrho)^2}\cdot\frac{W^{SO}}{W^{*}} \;=\; \left(\frac{1+2\varrho}{1}\right)^2\cdot\frac{\gamma^2}{(\gamma+2\phi)^2} \;=\; 1, \tag{22}$$

so in the symmetric case the Nash equilibrium is efficient; welfare losses arise only from asymmetry in knowledge.

10 Graphical Analysis

10.1 Agent interaction diagram

Figure 1 depicts the three-agent interaction structure.

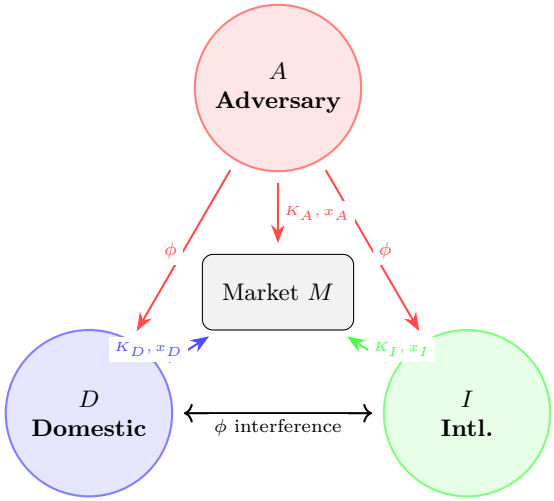

Figure 1: Three-agent MAID interaction structure. Arrows represent market participation weighted by fuzzy knowledge and impersonation level. Double-headed arrow between D and I indicates mutual interference. The adversary A introduces one-sided interference on both D and I.

10.2 Best-response functions

Figure 2 plots the best-response function of agent D as a function of K_D, for two levels of interference ϱ.

10.3 Equilibrium surface

Figure 3 illustrates the equilibrium impersonation of D as a function of (K_D, ϱ), holding $K_I = K_A = 0.5$.

11 Discussion

Robustness. The linear-payoff-plus-quadratic-cost structure is standard in industrial organisation [10] and yields tractable comparative statics. Generalising to n agents preserves the matrix structure, with the coefficient matrix becoming $\gamma\mathbf{I} + \phi(\mathbf{1}\mathbf{1}^{\top} - \mathbf{I})$, whose inverse is known in closed form.

Dynamic extensions. A repeated MAID game admits a folk-theorem characterisation: for sufficiently patient agents ($\beta \to 1$), the social optimum (20) can be sustained as a subgame-perfect equilibrium. However, with fuzzy knowledge, enforcement requires public monitoring, which may be unavailable when $K_j < 1$.

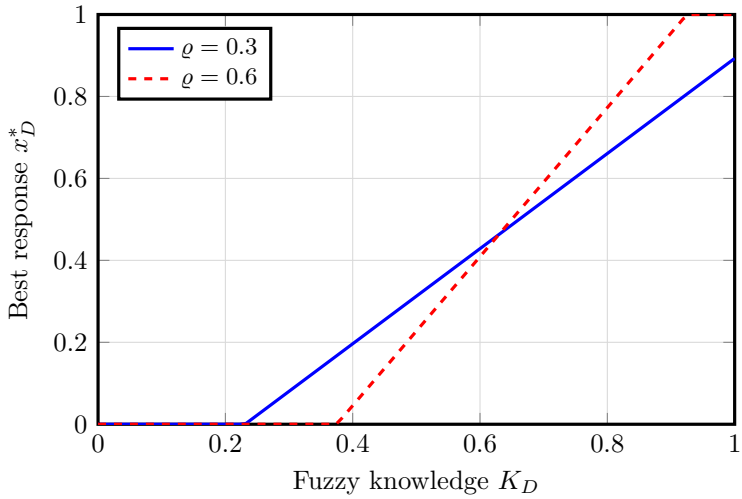

Figure 2: Best-response function $x_D^*(K_D)$ for $K_I = 0.6$, $K_A = 0.4$, $\mu/\gamma = 1$. Higher interference ratio ϱ steepens the slope and shifts the zero-impersonation threshold rightward.

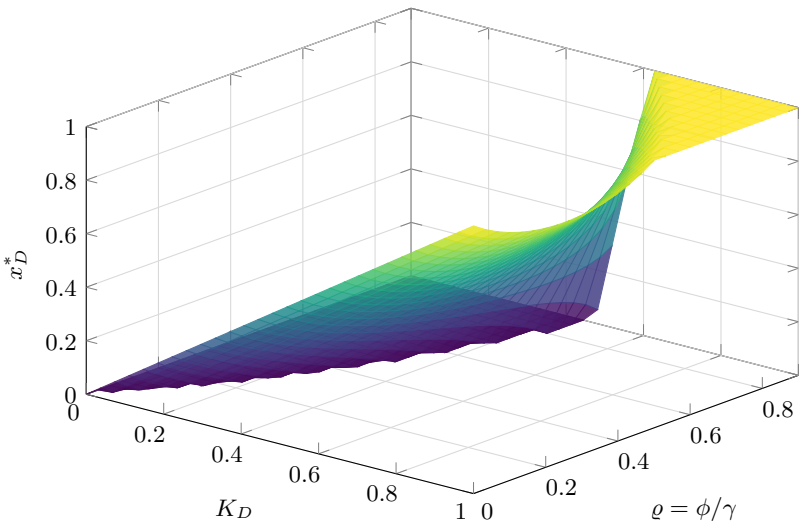

Figure 3: Equilibrium surface $x_D^*(K_D, \varrho)$ with $K_I = K_A = 0.5$. The surface flattens as $\varrho \to 1$ (collapse) and rises steeply with K_D when ϱ is small.

Regulatory implications. The Pigouvian tax in Corollary 9.1 requires a regulator to observe equilibrium impersonation levels. In practice, the regulator may only observe noisy proxies. Under measurement error $\hat{x}_j = x_j^* + \nu_j$ with $\nu_j \sim \mathcal{N}(0, \sigma_\nu^2)$, the optimal tax involves a shrinkage factor $1/(1 + \sigma_\nu^2/\mathrm{Var}(x_j^*))$.

12 Conclusion

We have introduced and solved the MAID model, characterising the unique Nash equilibrium of a three-agent impersonation game under fuzzy knowledge. The key results are: (i) a closed-form equilibrium formula with transparent comparative statics; (ii) strategic substitutability across all agent pairs; (iii) an adversarial knowledge wedge that suppresses cooperative participation even at low adversary knowledge levels; (iv) a collapse threshold at $\varrho = 1$ where the equilibrium degenerates; and (v) a Pigouvian tax that restores efficiency.

The model bridges game theory, fuzzy-set theory, market microstructure, and adversarial machine learning into a unified framework. Future work should address: dynamic MAID games with learning; heterogeneous interference structures; and empirical calibration to financial order-flow data.

References

[1] Crawford, V. P. and Sobel, J. (1982). Strategic information transmission. *Econometrica*, 50(6):1431–1451.

[2] Dubois, D. and Prade, H. (1988). *Possibility Theory: An Approach to Computerized Processing of Uncertainty*. Plenum Press, New York.

[3] Gal, Y. and Ghahramani, Z. (2016). Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, pages 1050–1059.

[4] Goodfellow, I. J., Shlens, J., and Szegedy, C. (2015). Explaining and harnessing adversarial examples. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

[5] Grossman, S. J. (1981). The informational role of warranties and private disclosure about product quality. *Journal of Law and Economics*, 24(3):461–483.

[6] Hansen, L. P. and Sargent, T. J. (2001). Robust control and model uncertainty. *American Economic Review*, 91(2):60–66.

[7] Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6):1315–1335.

[8] Nesterov, Y. (2013). *Introductory Lectures on Stochastic Optimization*. Springer, Berlin.

[9] Roughgarden, T. (2010). *Algorithmic Game Theory*. Cambridge University Press, Cambridge.

[10] Tirole, J. (1988). *The Theory of Industrial Organization*. MIT Press, Cambridge, MA.

[11] Zadeh, L. A. (1965). Fuzzy sets. *Information and Control*, 8(3):338–353.

Glossary

MAID
Multi-Agent Impersonation with Fuzzy Knowledge; the three-agent game studied in this paper.

Impersonation level (x_j)
A scalar in $[0, 1]$ measuring the intensity with which agent j misrepresents its identity or intentions; $x_j = 0$ is full authenticity.

Fuzzy knowledge (K_j)
A possibility-theoretic scalar in $[0, 1]$ indicating how accurately agent j observes the true market state; $K_j = 1$ is perfect knowledge.

Interference ratio (ϱ)
The dimensionless ratio ϕ/γ measuring how large cross-impersonation costs are relative to own impersonation costs.

Knowledge wedge
The suppression of a cooperative agent's equilibrium impersonation caused by the adversary's activity; formalised in Proposition 4.1.

Collapse threshold
The critical value $\varrho = 1$ at which the equilibrium matrix becomes singular and unconstrained equilibrium impersonation diverges.

Strategic substitutability
A property of best-response functions where increasing one agent's action decreases rivals' optimal responses; characterised by $\partial \mathrm{BR}_j / \partial x_k < 0$.

Price of anarchy (PoA)
The ratio of social welfare at the social optimum to social welfare at the Nash equilibrium; a measure of efficiency loss from non-cooperative behaviour.

Pigouvian tax A per-unit tax on impersonation equal to the marginal external cost imposed on other agents; decentralises the social optimum (Corollary 9.1).

Projected gradient descent
An iterative optimisation algorithm that follows the negative gradient and projects iterates onto a feasible set; used in Algorithm 1 to compute the constrained MAID equilibrium.

Value-at-Risk (VaR)
The loss not exceeded with a given probability (here 99%) over one period; increasing in impersonation level and knowledge uncertainty.

GMM Generalised Method of Moments; a semi-parametric estimator for structural parameters using moment conditions derived from the equilibrium FOCs.

The End